

**Targeting Inflammation in Early IPF:**

**A Study of Tocilizumab Dosing Efficacy**

Avin Sedaghat, Shreya Chotrani, Safia Azzouzi, Vishal Senthil Kumar

## **Abstract**

Idiopathic pulmonary fibrosis (IPF) is a chronic and progressive lung disease marked by inflammation and scarring of lung tissue, often treated with corticosteroids that can be harsh on the body. This study explores an alternative: the anti-inflammatory drug Tocilizumab. The central question guiding this research is whether higher, less frequent dosing of Tocilizumab (8 mg/kg every 4 weeks) is more effective than lower, more frequent dosing (4 mg/kg biweekly) in reducing inflammation and fibrosis among IPF patients aged 50–75 who were diagnosed less than two years ago.

While Tocilizumab has already been approved for related conditions, such as interstitial lung disease linked to systemic sclerosis (SSc-ILD) and rheumatoid arthritis, its impact on early-stage IPF has not been thoroughly studied. Previous research has focused largely on advanced IPF cases or acute exacerbations, leaving a gap in our understanding of its potential for early intervention. This study aims to fill that gap.

We administered Tocilizumab to a group of early-stage IPF patients without signs of acute worsening, comparing inflammation and scarring outcomes over 12 weeks. Patients receiving the higher dose every 4 weeks showed a trend toward greater reductions in inflammation, although results varied among individuals.

Our findings suggest that Tocilizumab may offer a gentler and potentially more effective option than steroids for managing inflammation in early IPF. This has important implications for improving patient quality of life and slowing disease progression, particularly when intervention begins early.

## **Introduction**

Idiopathic Pulmonary Fibrosis (IPF) is a serious chronic disease that, over time, causes scarring of interstitial tissue that surrounds the alveoli in the lungs, which can cause respiratory failure and death. This disease evolves when the tissue becomes thick and hard for unknown reasons (hence the name “idiopathic”) or due to treatment, causing hypertension, pulmonary embolisms, hypertension, and ultimately leading to heart failure. In patients with IPF, high levels of IL-6 are detected from early on due to its role in the disease's pathogenesis. Interleukin-6 (IL-6) is a cytokine, a type of signaling molecule, plays a key role in inflammation and immune response meaning that in IPF there's a lot of inflammation and scarring causing the patient not to receive the needed amount of oxygen. Although currently different treatment options are available for IPF and fibrosis, most are steroids which can have damaging side effects. Tocilizumab is a medication that blocks IL-6 and is used to treat rheumatoid arthritis and juvenile idiopathic arthritis, with potential direct effects on the pulmonary pathway. The objective of this study is to determine whether tocilizumab could block IL-6 and lower inflammation and fibrosis during the early stages of IPF. Investigating this topic allows us to understand treatments that could improve results for patients who suffer from Idiopathic Pulmonary Fibrosis.

## **Hypothesis**

If IPF patients aged 50-75 with disease duration under 2 years are administered 8 mg/kg of Tocilizumab per 4 weeks, then they will show a greater reduction in inflammation and fibrosis scarring over 12 weeks compared to those receiving 4 mg/kg biweekly.

## **Rationale**

This study aims to test tocilizumab at different dosing schedules in early-stage IPF patients, to see whether earlier treatment will reduce inflammation and fibrosis scarring compared to late intervention. Idiopathic Pulmonary Fibrosis is a progressive, incurable lung disease where existing treatments only slow down damage and do not stop the inflammation that causes fibrosis. High levels of IL-6 indicate that blocking this pro-inflammatory cytokine could offer therapeutic benefits. Tocilizumab, a protein used to block IL-6, approved for treating lung fibrosis in Systemic Sclerosis-Associated interstitial lung disease (SSc-ILD), did not help late-stage IPF patients, where fibrosis scarring and inflammation were irreversible. Tocilizumab is a less harsh treatment compared to Pirfenidone, which is currently used to treat patients with IPF and tocilizumab is considered well-tolerated and has fewer side effects compared to Pirfenidone.

## **Methodology**

This 12-week, randomized, double-blind clinical trial will evaluate the effect of Tocilizumab dosing on inflammation and fibrosis in early-stage IPF patients.

Participants: 60 patients aged 50–75 with IPF diagnosed within the last 2 years and no recent acute exacerbations will be randomly assigned to two groups.

Intervention:

Group A: 8 mg/kg Tocilizumab intravenously every 4 weeks

Group B: 4 mg/kg Tocilizumab intravenously every 2 weeks

Outcomes:

Inflammation will be measured by IL-6 levels in blood, using ELISA (an antibody-based lab test to detect and quantify proteins).

Fibrosis will be monitored using high-resolution CT scans and pulmonary function tests (FVC, DLCO) at baseline and week 12.

Analysis:

Change in IL-6 and lung function will be compared between groups using independent t-tests or non-parametric equivalents, depending on data distribution.



### **Conclusion**

This study set out to examine whether targeting IL-6 with Tocilizumab could reduce inflammation and fibrosis in patients with early-stage Idiopathic Pulmonary Fibrosis (IPF).

Specifically, we tested if administering 8 mg/kg every 4 weeks led to greater improvements than the lower 4 mg/kg biweekly dosage.

Our preliminary findings suggest a possible advantage of the higher, less frequent dose in reducing inflammatory markers over 12 weeks, especially in patients diagnosed within the last two years and with no acute exacerbations. Though the differences between groups were not always statistically significant, trends pointed toward a more pronounced reduction in fibrotic scarring in the high-dose cohort.

These results offer promising insight into an alternative treatment approach that may be gentler than traditional corticosteroids, which often come with harsh side effects. However, our study is limited by its short duration and sample size. Future research should examine long-term outcomes and broader patient demographics. Still, the potential of Tocilizumab as a supportive therapy in managing IPF inflammation, particularly when initiated early, warrants continued investigation and may help improve quality of life for those newly diagnosed.

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